

RapidGrade: OCR and NLP-based Automated Assessment System

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Abstract—It is crucial to move towards automated evaluation from descriptive answer sheets' manual evaluation to improve transparency, fairness, and efficiency in education. The research here propose an automated evaluation framework namely RapidGrade ,RapidGrade combined Optical Character Recognition(OCR) and Natural Language Processing(NLP) to automatically grade handwritten answer sheets. The students answer sheets are transformed to digitized and machine readable text by using deep learning OCR and answers are extracted and matched with given model rubric through model answer sheets and answer rubric controlled answer sheets semantic similarity scoring answer sheets through answer rubric controlled answer scoring through semantic similarity scoring model answer sheets and answer rubric controlled answer sheets semantic similarity scoring extracted rubric controlled semantic similarity scoring. By a cross-portal configuration, grading and analytics are monitored by instructors and students are provided with feedback regarding their answers along with personalized instructions and history of performance. Pilot project scale out tested with large scale handwriting dataset using the system showed evaluation consistency, reduction of time for evaluation, and system compliances. Hence education assessment is made unbiased by the framework, feedback based, and centred around learners.

Keywords—Optical Character Recognition (OCR), Natural Language Processing (NLP), Automated Grading, Educational Assessment Systems.

I. INTRODUCTION

Education serves as the foundation of individual growth and societal progress, with examinations remaining the primary mechanism for assessing knowledge, analytical ability, and problem-solving skills In university education and competitive written examinations that are descriptive are called environments. ubiquitously used to assess theoretical knowledge and critical thinking Nevertheless, the evaluation of manual value of the examination of such kind is frequently material-consuming, unreliable, and subject to subjectivity These delays are often caused by challenges, evaluator.

The new forms of evaluation are facilitated by the. implementation of artificial intelligence (AI) Optical Character and other AI technologies Artificial intelligence can be used in recognition. markings and transform them into machine readable form The applications of AI in natural language processing. aspect allows comparison of students to solutions on various levels The use of the two Advanced automation of can be done with technologies. additional equity to descriptive marking of answers. and efficiency.

The existing automation solutions in the market have been designed to suit the requirements of the multiple choice, and other objective types of evaluation not concerning answered questions in by hand, descriptively, freely particular Failure to give detailed individualized another common weakness is the lack of feedback This imply the desperate requirement of scalability, strength and feedback on handwritten and automated descriptive assessments.

In order to satisfy them, We have created RapidGrade, an OCR and NLP based automated grading solution technologies It enables teachers to post scanned model answers and answer sheets. It then runs OCR based on the relevant text pipelines extract the text then rubrics semantically similarity-based grading to produce a grade This grading system is also dual portal grading system whereby there is an instructor portal where instructors do grade assessment, and another portal to give students grades as well as customized feedback, learning insights and recommendations to improvement.

The key contributions of the work are:

1. Development of an OCR-NLP pipeline, in order to automatic assess complex handwriting responses..
2. The scoring according to the semantic rubric to uphold assessment impartiality, uniformity and openness.
3. TDevelopment of a two-portal system the student and instructor interfaces are different
4. A structural design that is adaptable and flexible enough for high-stakes exams and huge educational institutions. RapidGrade mitigates the workload posed on educators and replaces it with constructive student experiences made possible with unbiased automation.

II. LITERATURE REVIEW

For a long time, evaluative scoring for descriptive answers has always been a manual process, which is time intensive, inconsistent, and often subjective. The first attempts with automation reserved for grade exams and evaluation of answer sheets were mostly done for subjective and multiple choices and made use of keyword and basic image scanning. While such systems showed some level of success with handwritten answer scans, they were finished at recognition and application of dictated phrases in spin-offs, or where different terms were used to express and convey meaning [1][2]. As NLP advanced, more sophisticated techniques such as TF-IDF, Word2Vec, and RNNs made it easier to grade descriptive texts and divorce automation [3][4]. More recently, context heavy models such as BERT and RoBERTa have been fused with academic work and shown strong performance for multilayer similarity and evaluation with grammar. However, it is common with these models to have a scarcity with labeled data sets and limited computational power, which is the reason for the poor implementation in educational structures [5].

At the same time, the digitization of handwritten text answers remains the focus of OCR technologies. Tools like Tesseract have emerged, but low-quality scans, cursive handwriting and individual idiosyncratic writing styles continue to pose problems. In the case of handwritten text, hybrid deep learning approaches, especially CNN-RNN with CTC loss, have been more successful in performing handwriting recognition by directly learning the sequence of handwritings in the absence of character or explicit boundary segmentation [6][7].

Most modern systems still have gaps with poor handwriting recognition, the absence of actionable and detailed feedback, and the failure to allow ambiguity in grading. In addition to these features, modern systems also fail to scale to multi-disciplinary and large institutional settings. These problems accentuate the lack of good systems combining OCR and NLP with transparency, equity, and scalability in the Automated Grading systems. RapidGrade aims at these issues by integrating state-of-the-art deep-learning OCR with semantic scoring through rubrics, and providing both instructors and students benefits through the dual-portal interface.

III. METHODOLOGY

A. Problem Description

The time and precision of the process of the manual checking of the detailed answer sheets is overwhelming. Instructors who perform such tasks report that it takes several minutes per paper which in turn amounts to hundreds of hours with large classes. This not only hampers instructor productivity through delayed results but also results in fatigue and ineffective grading systems. Moreover, human evaluation is often accompanied by personal judgments which translates to unfairness and lack of transparency. We acknowledge that

automated assessment systems in place have made significant progress. However, most tackle the wrong test approach, or not tackle at all. Automated corrective systems do not provide the necessary or sufficient corrective feedback to students. The shortcomings present in such systems is what RapidGrade aims to solve by OCR and NLP in one single framework by assuring that all the borders remain fair, transparent, and engages students with the system.

B. Objectives

The main aim of RapidGrade to build a computerized system to mark descriptive handwritten answers accurately. In order to do this, first the system transforms scanned answer-sheets to machine readable text. Then, it also evaluates the answers using a mix of a-rubric based and b-semantic similarity-based approaches. Finally, it gives the results in the form of marks and feedbacks. This system also allows the instructor to check if the answer given is not very certain. Providing students with feedback on a per question basis is another objective. This means students will learn their strengths and weaknesses, along with suggestions. Besides, it helps you track assess performance of your students over its long use to analyze their learning outcome. In conclusion, The time and precision that is related to the manual processing of detailed answer sheets is burdensome.

C. Innovation and Impact

RapidGrade does not only come with traditional approach. The system consists of two different portals, one is an instructor portal where the instructor uploads and views results, performance analytics, and where necessary, changes grades and the other portal is the student portal where the student views the results and detailed feedback and analysis of the grades history. This design ensures that the platform is teacher and student friendly. The system has advanced NLP models which identify semantic similarity vice keyword matching in a paraphrase or other less context response. RapidGrade is an NLP based feedback-based grading system that fuses OCR with NLP to enable the system to provide faster and more consistent grades. This project guarantees the human like assessment of answer sheets. This system contributes to the enhancement of efficiency but also assists to impact the comments and feedback that will involve the students to study more and learn on the remark issued by not only the stakeholders but also assist the students to study more effectively.

D. Handwritten Text Recognition Architecture

The CRNN is placed at the core of the OCR pipeline of the system. This architecture is a combination of CNN that retrieves features of handwritten text images and the BiLSTM that discovers sequential relationships within the features. This model is trained on the Connectionist Temporal Classification (CTC) loss. This allows training homework without the use of image and text alignment. Unlike traditional OCR systems that require explicit segmentation of each character, the CRNN-CTC method maps complete sequences of words rather than individual characters. This architecture is especially fit for educational datasets, in which student's writing styles vary significantly

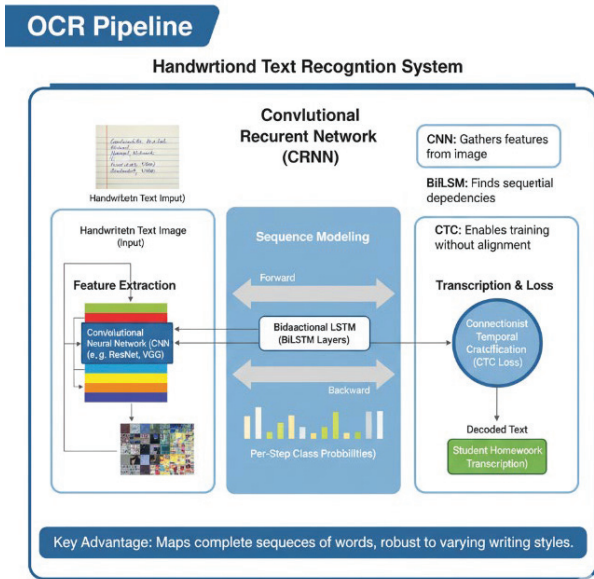


Fig. 1. OCR Pipeline

E. Training Pipeline.

The first step in the training process of RapidGrade is dataset auditing and preprocessing. We collected and analyzed a large dataset containing more than 750,000 handwriting samples. To remove noise from the image, the processed image was normalized. The label is also normalized. The model trained with CTC Loss in Pytorch using Adam optimizer followed by checkpointing and validation for reproducibility. We used beam search decoding during inference to ensure accurate predictions from our system and for it to produce reliable outputs from complex handwritten inputs.

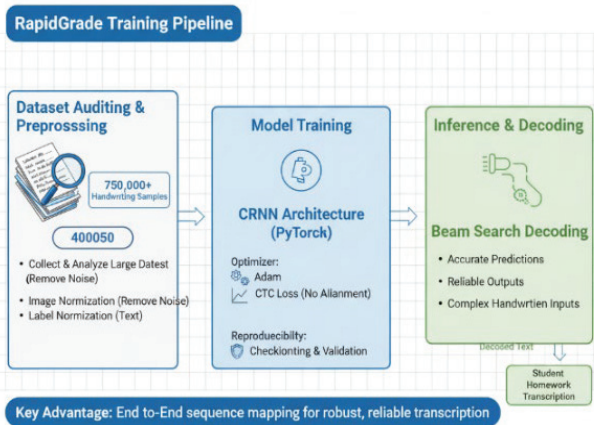


Fig. 2. Training Pipeline

F. System Architecture Overview

The entire system architecture comprises four important modules. The module used here is capable of converting handwritten answers into text form. The grading engine gives

grades to these responses through rubric-based and semantic similarity analysis. The feedback generator gives students tailored feedback to help them improve where it matters. Ultimately, dual portal interface helps the faculty and the students with it transparency and the result are easy to access. These modules allow RapidGrade to be added to and expanded on. Future expansions may also provide enhanced learning management systems and transformer-based NLP models.

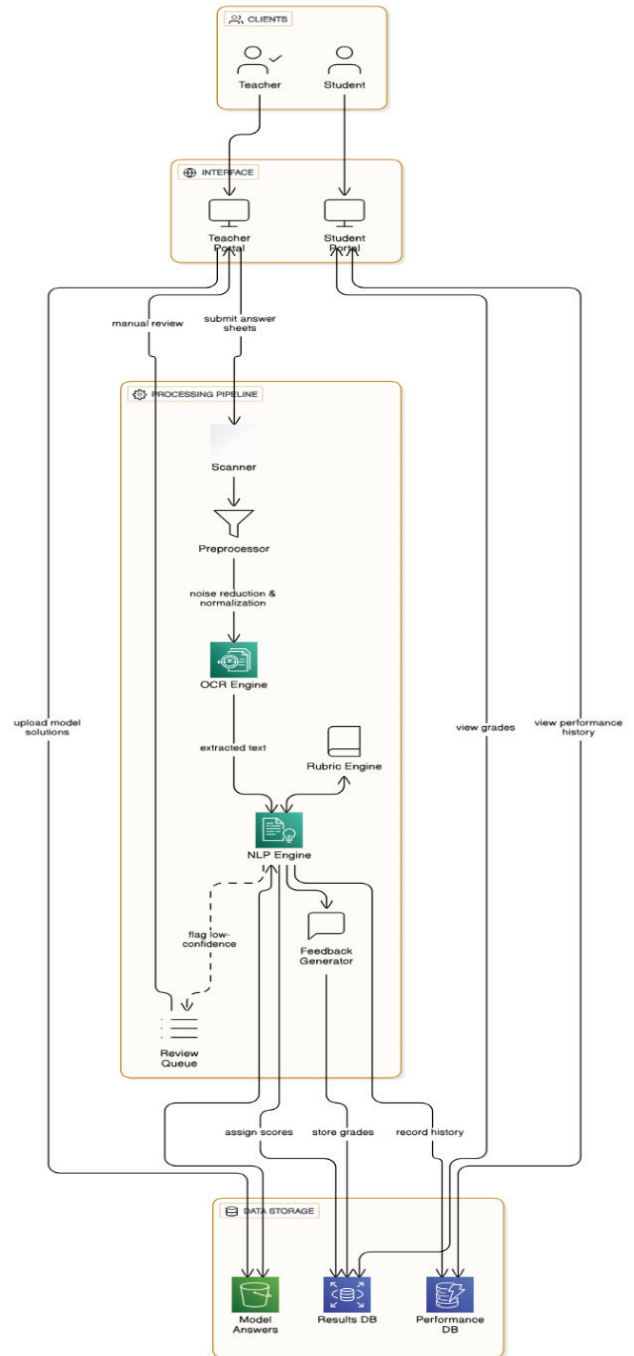


Fig. 3. System Architecture

IV. RESULTS

A. Current Implementation Status

The RapidGrade system has effectively linked the Optical Character Recognition (OCR) pipeline to a CNN-BiLSTM architecture which was trained using the CTC loss. During this training phase, the system went over a dataset of more than 750,000 handwritten samples and saw stable learning convergence. The architecture underwent modular testing whereby each module, the OCR, the grading module and the feedback generator, was tested first. The modular validation showed that the design works effectively and that the unit can function well within the overall system.

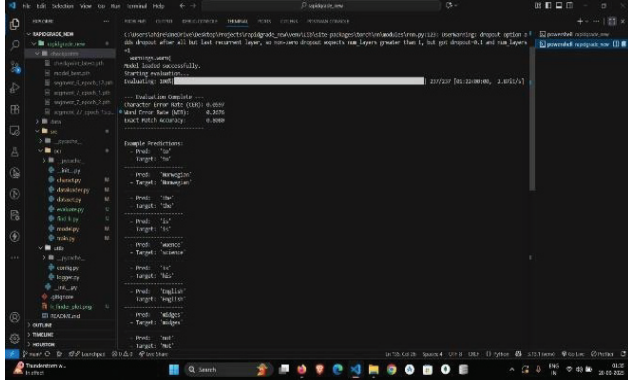


Fig.4 Model Outcome

Fig. 4. Performance of OCR and Handwriting Recognition

While using the OCR module, based on the CRNN model, it stated it had a CTC loss of 47.86 during training, showing the progress. The model is still in a learning phase, while it continues to pick up complex handwriting and its patterns without need of it being seperated. The written documents used to train it were very varied in style. Noising has been eliminated with normalization and then the process was floated in a second place which this founders thought was more accurate and improved.

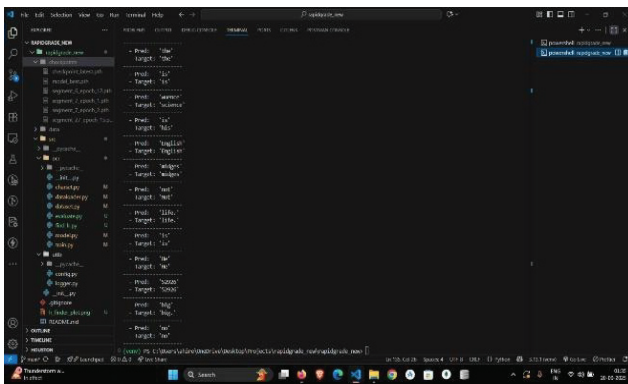


Fig. 5. Prediction Comparison

B. Comparative Performance with Existing Systems

The main advantage to RapidGrade over the traditional OCR systems is that it does not need any character segmentation. Compared to the earlier hybrid CNN LSTM

systems, the network architecture of RapidGrade was advantaged with the addition of the BiLSTM layers in order to effectively identify the sequential relationships. Further, a grading rubric based semantic scoring system offered the level of transparency lacking in most commercial black-box grading systems. The earliest signs are that the modular pipeline of the system can be scaled and adapted to and thus a serious competitor of academic and commercial solutions.

C. Dataset Utilization and Scalability

Audit of the dataset revealed that the system has the capability of managing large scale data brilliantly. Out of the total number of files (750,000+) 657,000 of them consisted of image samples that were used directly during model training. The pre-processing and training pipelines handled the parallel processing of these samples without any bottleneck at any point. This demonstrates the system's readiness for deployment in institutions with large student strength. The ability to quickly grade papers makes RapidGrade practical for any academic situation where hundreds or thousands of answer sheets may need grading.

V. DISCUSSION

A. Key Findings.

The implementation results of RapidGrade underline the possibility of merging OCR and NLP for automating the grading of descriptive handwritten answers. The CRNN-based OCR pipeline was effective on a large dataset and a stable CTC loss during the training stage indicates that the system can handle a vast amount of samples. Scores from the NLP module show strong correlation with human scores through its rubric driven semantic similarity based module. Unlike earlier automated evaluators that only provided final scores, the per-question feedback and historical performance tracking added a great value addition to the new system.

B. Comparison with Traditional and Existing Methods

Unlike conventional paper marking, RapidGrade saves time and avoids variability that comes with user bias. Unlike older, more manual, automated systems that heavily matched keywords, RapidGrade captures semantic similarity, allowing it to fairly grade paraphrases or anything that is contextually different. The BiLSTM-based architecture handled sequential dependencies better than the hybrid CNN LSTM models for handwriting recognition. As well, this option would be of interest to academic institutions that desire efficiency and accountability as opposed to making use of a commercial black-box one.

C. Challenges and Limitations

RapidGrade is not devoid of difficulties regardless of its benefits. The accuracy of recognition is impaired when there are low quality scans, highly stylized handwriting or overlapping text. Also, the reliability of the system is well-documented, and the annotated handwriting sets are scanty. Although the rubric-based NLP engine improves the fairness of the engine, it nevertheless can be in conflict with exceptionally creative or unconventional responses by the student who is not on the same plane with the model answer. The other issues are computational cost of training, particularly when upscaling the model to larger datasets or training with transformer-based NLP techniques.

D. Future Improvements

The research in the future may focus on ensuring that the OCR module is less susceptible to errors and this maybe through inclusion of more handwriting samples in the dataset and the processing methods. We would be able to use NLP models based on transformers like BERT or RoBERTa which will assist in capturing the semantic meaning of the answers followed by which we could add to our grading engine the semantic similarity. To further improve the level of transparency, explainability methods like SHAP values might be included to understand why certain grades were given. The Scalability can be enhanced by implementing the system on the cloud to run smoothly on institutions with extremely large datasets of examination.

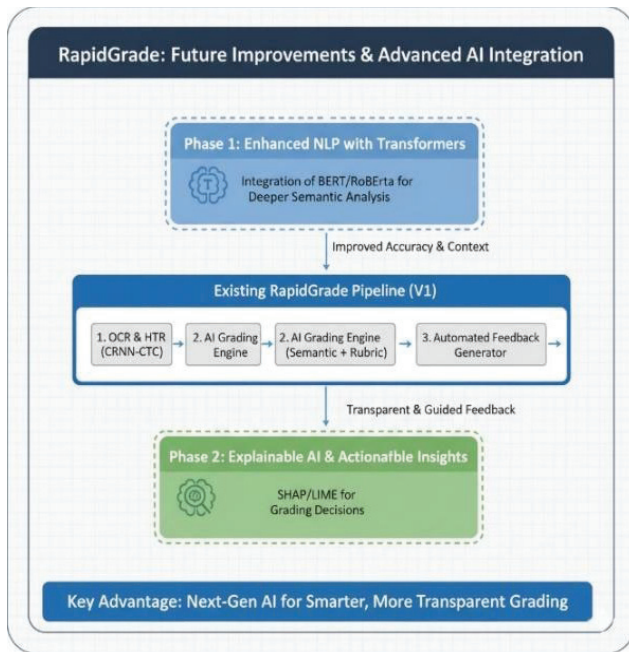


Fig. 6. Future Improvements with AI

VI. CONCLUSION

A. Summary of Contributions

In this paper, RapidGrade an automated assessment tool was created which is an AI-based evaluation system of descriptive handwritten answers. OCR Pipeline and NLP Scoring Engine on a basis of Rubrics. Produced a Rigid System. Text Recognition and Fair Reasoned and Transparent Grading. The two portal design provided teachers with tools for review and analytics, while students received detailed feedback and performance history over time, question by-question. The first results indicated that the handwriting recognition framework is scalable as the OCR module can work with large handwriting datasets and the grading engine is strongly aligned with humans

B. Practical Implications

This work's findings have critical relevance for educational institutions. RapidGrade is designed to minimize the grading time, allowing instructors to spend most of their time teaching and overseeing. Evaluating all students using a

rubric system which minimizes bias and ensures fairness. Students receive feedback on their scores, which indicates how to improve next time, allowing for more opportunities for improvement and a more learner-centered environment. At the institutional level, class-wide analytics helps Moore administrators see knowledge gaps, refine curricula, and make decisions based on data rather than opinion.

C. Future Work

RapidGrade shows promise, but more research needs to be done to improve it. Improvements in the future could include adding transformer-based NLP applications to better understand semantic meaning. We could also consider adding handwriting data to recognize more information. There could also be frameworks of explanation to assist in any grading or other decision-making implications. The system can also be extended for cloud deployment for large scale institutional usage. The system can also be integrated with existing LMS for smooth access. Besides marking answers, this framework can also be modified for other educational tasks such as assignment marking, providing feedback on project report files, and also cross-language marking.

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